

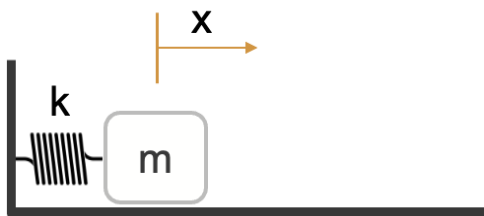
PHY-275 Fall 2026 Assignment #2 due 9/4/26 9:00pm

You will receive 1 credit upon complete all the tasks.

Task #1: Derive system of equations on your own

In class we did this:

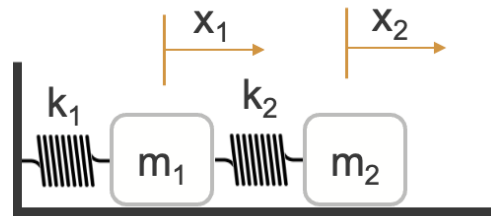
Single harmonic oscillators: Double harmonic oscillators case A:



$$\dot{x} = v$$

$$\dot{v} = -\frac{k}{m}x$$

$$\begin{bmatrix} \dot{x} \\ \dot{v} \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -\frac{k}{m} & 0 \end{bmatrix} \begin{bmatrix} x \\ v \end{bmatrix}$$



$$m_1 \ddot{x}_1 = -k_1 x_1 + k_2 (x_2 - x_1)$$

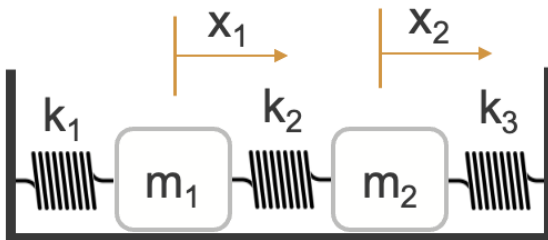
$$m_2 \ddot{x}_2 = -k_2 (x_2 - x_1)$$

$$\begin{bmatrix} \dot{x}_1 \\ \dot{v}_1 \\ \dot{x}_2 \\ \dot{v}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ -\frac{k_1+k_2}{m_1} & 0 & \frac{k_2}{m_1} & 0 \\ 0 & 0 & 0 & 1 \\ \frac{k_2}{m_2} & 0 & -\frac{k_2}{m_2} & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ v_1 \\ x_2 \\ v_2 \end{bmatrix}$$

**Assuming: no friction, no air resistance*

Now do this:

Double harmonic oscillators case B:



To proof your effort, write down the steps: free body diagram, equation of motion, and then the matrix. Feel free getting any help you need. But making sure you can eventually work out the whole thing on your own. **This task will show up in mid-term.**

Task #2: Write a brief summary of the research paper you did in assignment #1, task #5. One slide, and answer the following four questions:

Q1: Main scientific questions (for science)? or Practical challenges (for engineering)?

Q2: Current understanding/progress (need to list the mainstream opinions as bullet points)? Knowledge gap (for science)? or Technology barrier (for engineering)?

Q3: Innovation and uniqueness?

Q4: What can you learn from the literature?

Here's one example from my previous research group:

Q1: Main scientific questions (for science)? or Practical challenges (for engineering)?

- How activity induced new spine form new synapse
- Spine formation and new synapse connection in organoids memory

Q2: Current understanding/progress (need to list the mainstream opinions as bullet points)? Knowledge gap (for science)? or Technology barrier (for engineering)?

- How newly formed spine turns to synapse and incorporate into the neural network
- No behavior or animal experiment to prove their finding is related to memory (2009 nature paper)

Q3: Innovation and uniqueness?

- Define a score for spine
- Found part of newly formed spine could turn to pre-existing type and incorporate into the network

Q4: What can you learn from the literature?

Established biological and neurological evidence for our organoid memory work:

- New spine formation is related to memory formation
- Time for spine formation and synapse formation
- Electrical stimulation could induce spine change

Welcome to use AI's help. But validate and polish the writing on your own. You know there'll be follow-up questions regarding this task huehuehue.